

Upcoming Dates

- **Due Today:**
 - HW 13
 - Composite video note
- **Thursday: Module 6 Lab Worksheet**
 - Proficiency testing #5
- **Tuesday, May 5 – Module 6 Videos Due**
 - Bring Composite To Class
- **Friday: Watch Module 6 videos, Post Lab worksheet due**

Final Exam

- **Friday, May 15, 7 AM – 9 AM**

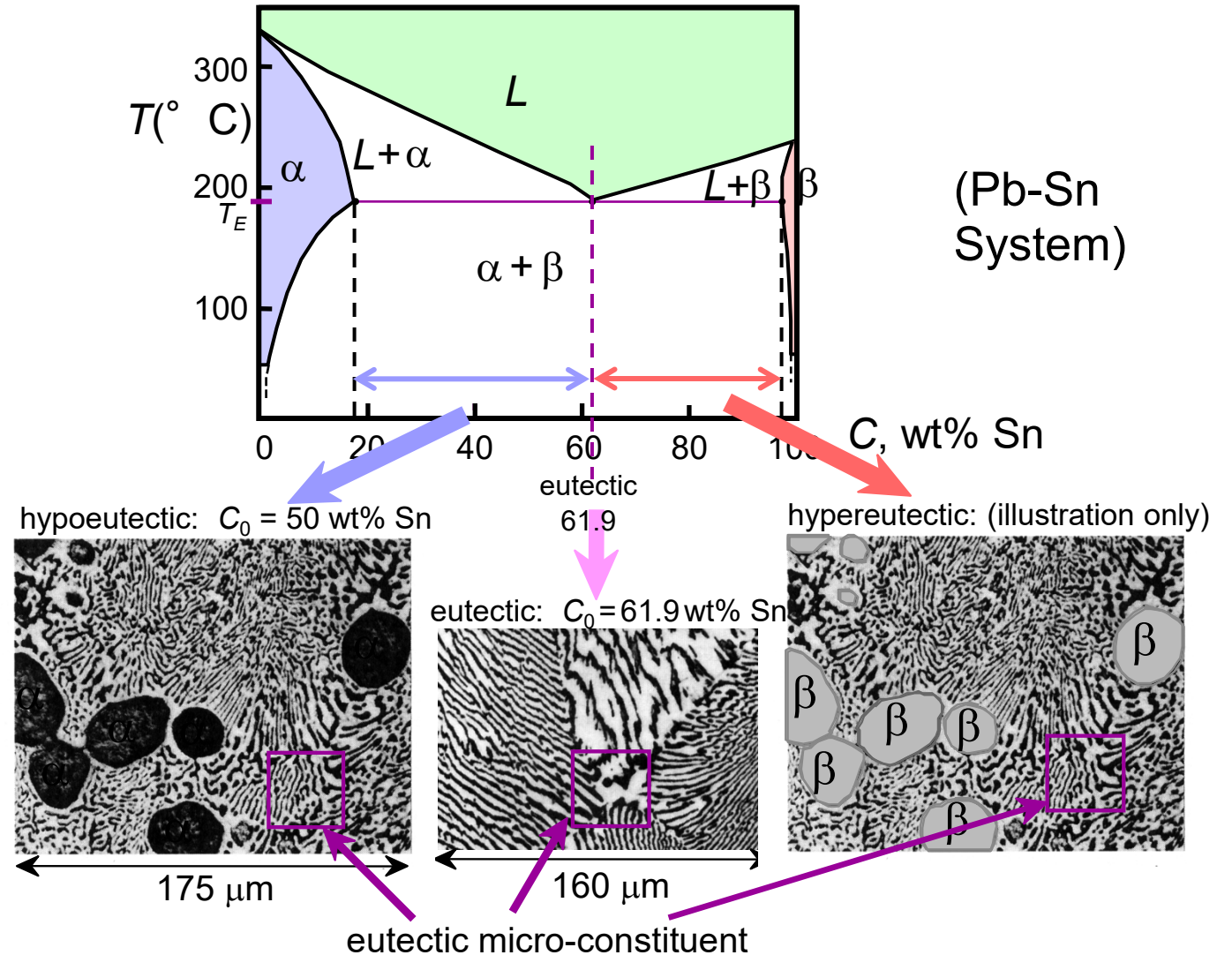
Lecture 16 – Maximum Solubility

The maximum solubility for Sn in Pb is:

α particles are mostly Pb...
18.3% at $T=183^{\circ}\text{C}$

The maximum solubility for Pb in Sn is:

β particles are mostly Sn...
2.2% at $T=183^{\circ}\text{C}$



Iron-Carbon (Fe-C) Phase Diagram

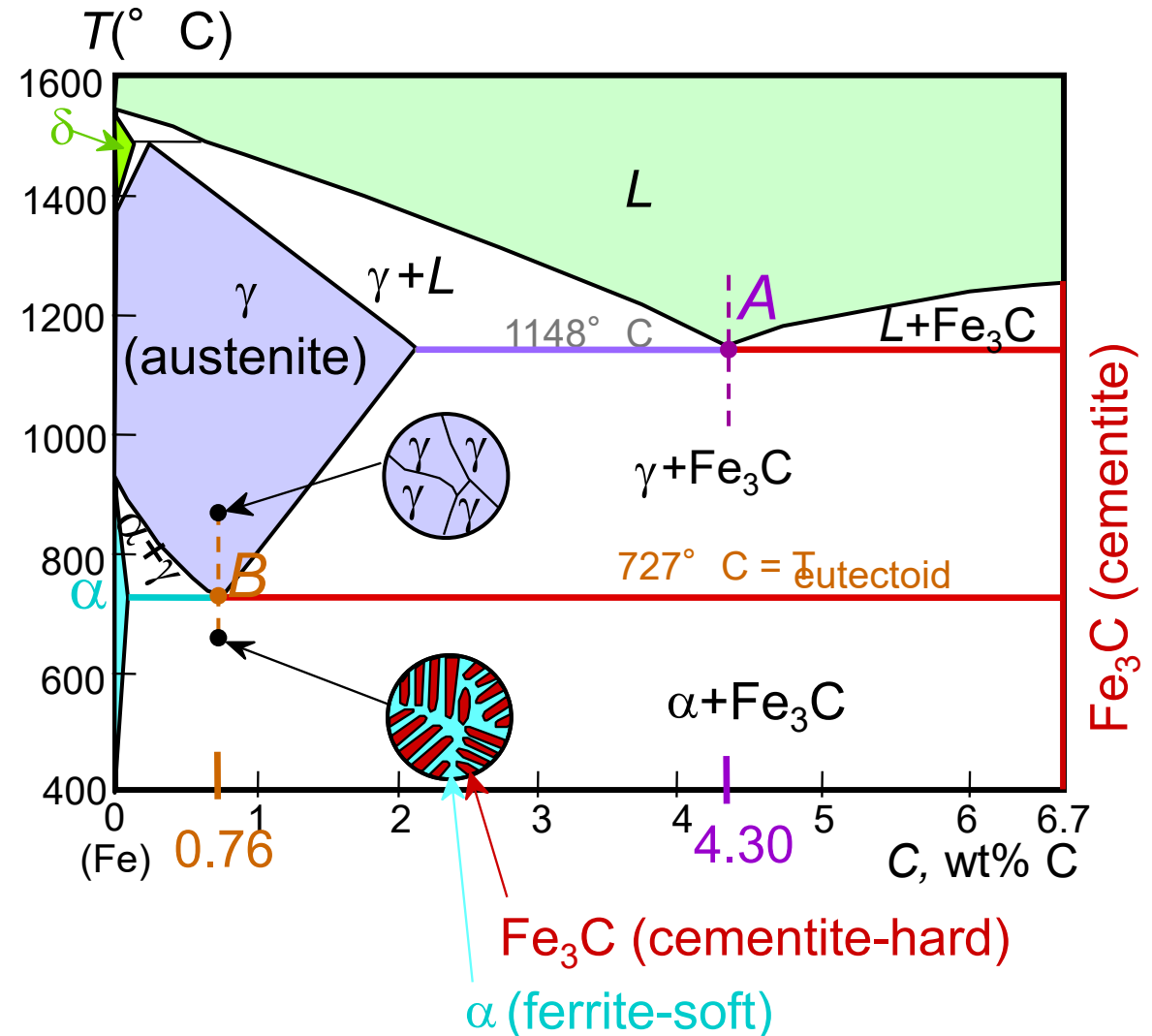
Carbon dissolved easily in the austenite

Max solubility limit for carbon in austenite is:

2.14% at $T=1148^{\circ}\text{C}$

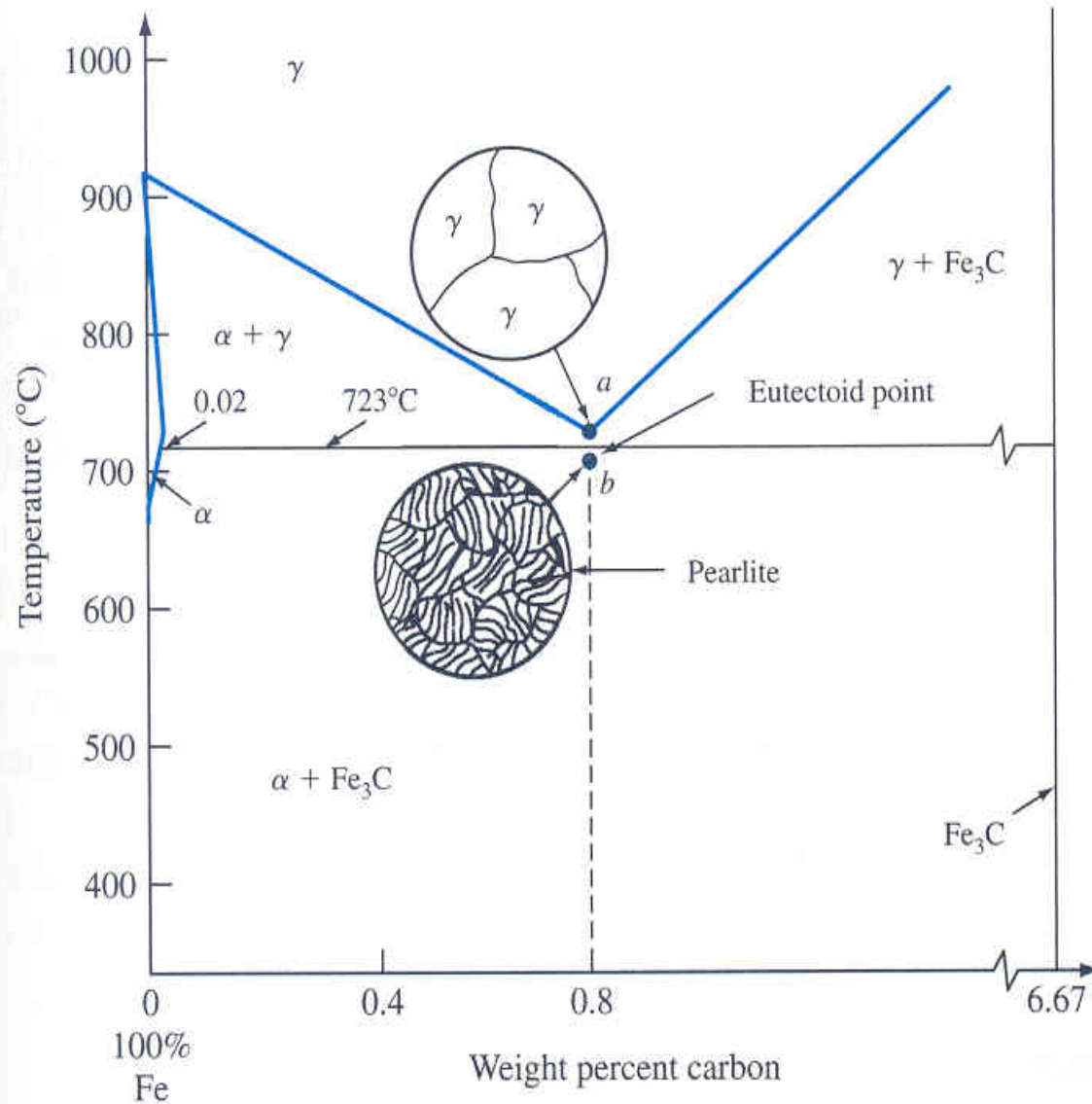
For ferrite, BCC, The max solubility for carbon is:

0.022% at $T=727^{\circ}\text{C}$

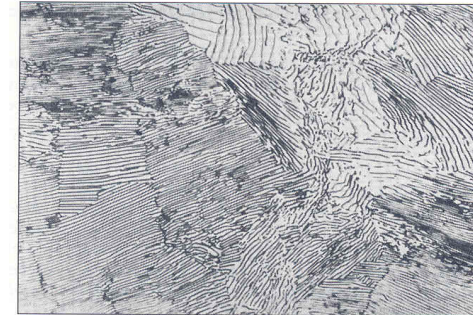


Slow Cooling of Plain Carbon Steel

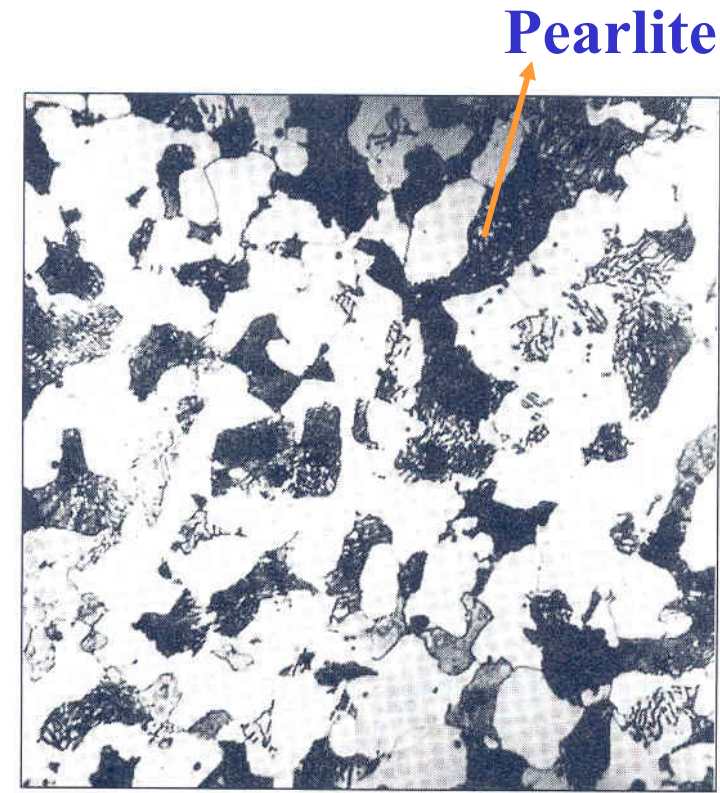
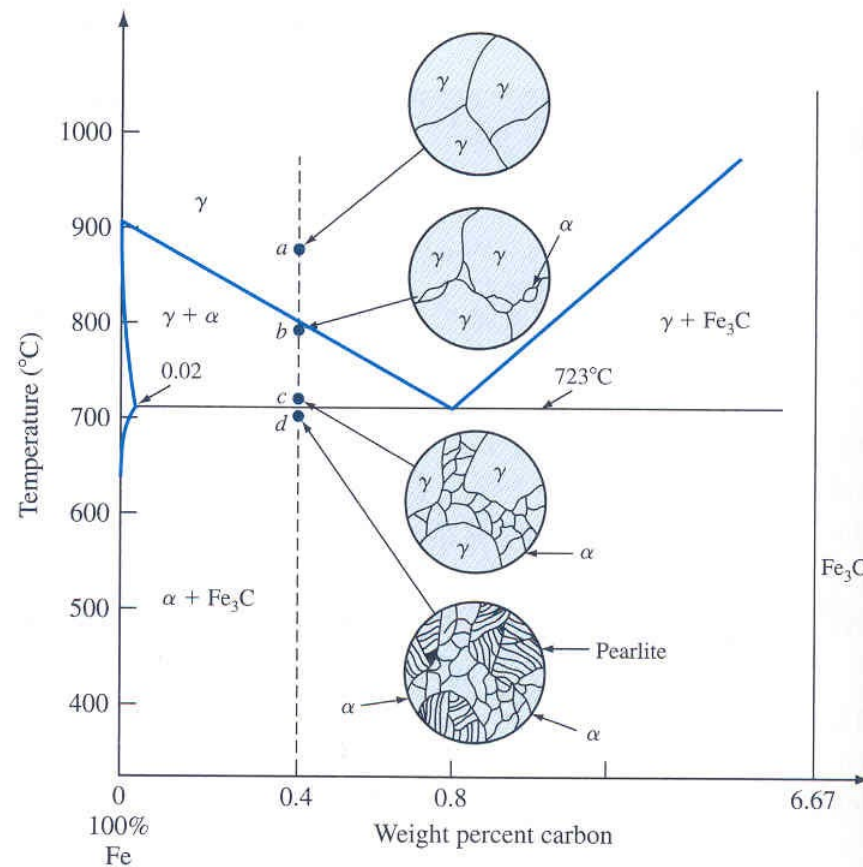
austenite--- pearlite if sufficient long time.



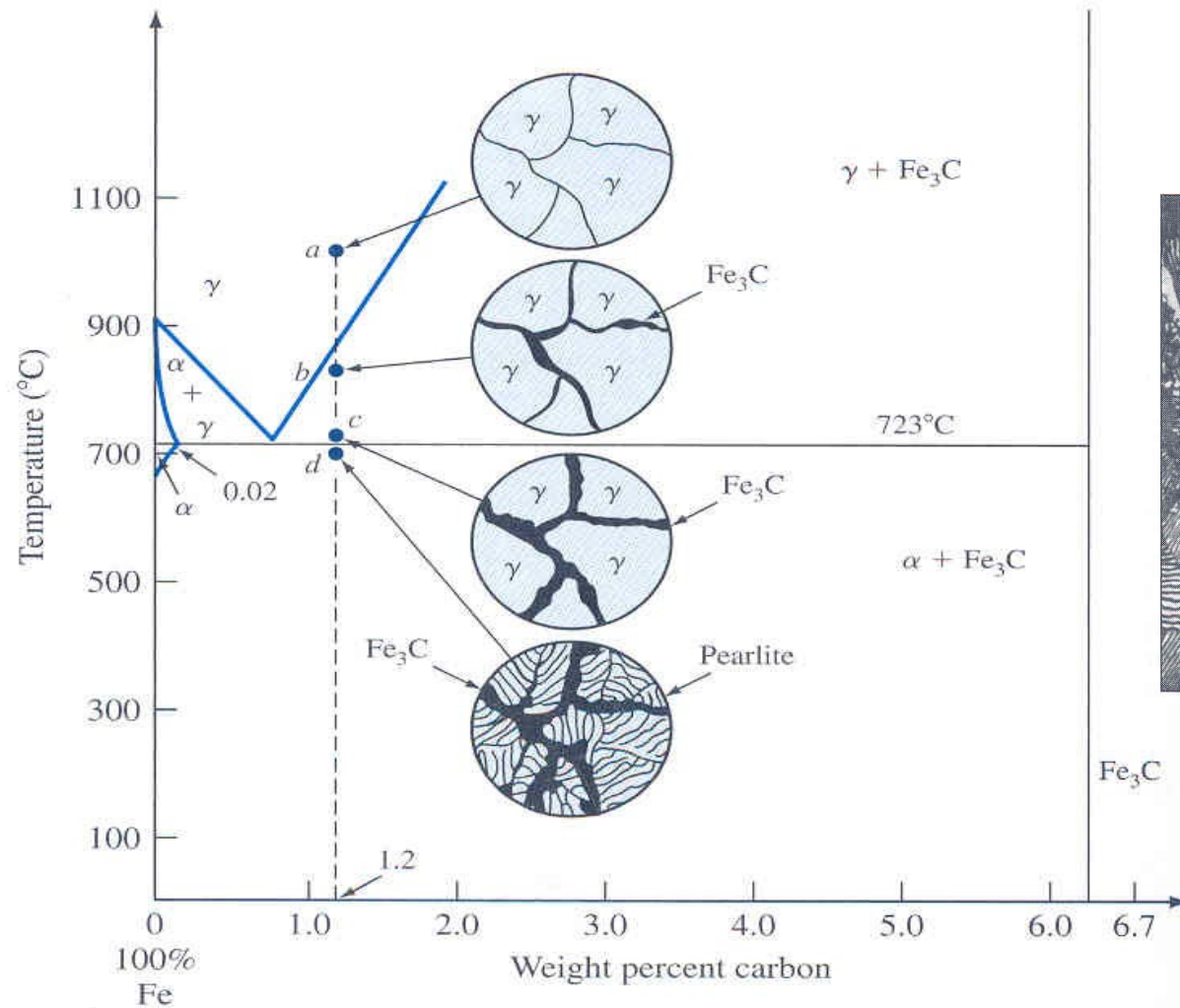
Pearlite



Slow Cooling of Plain Carbon Steel

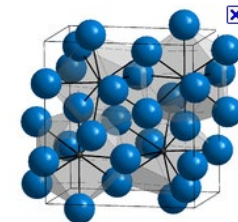
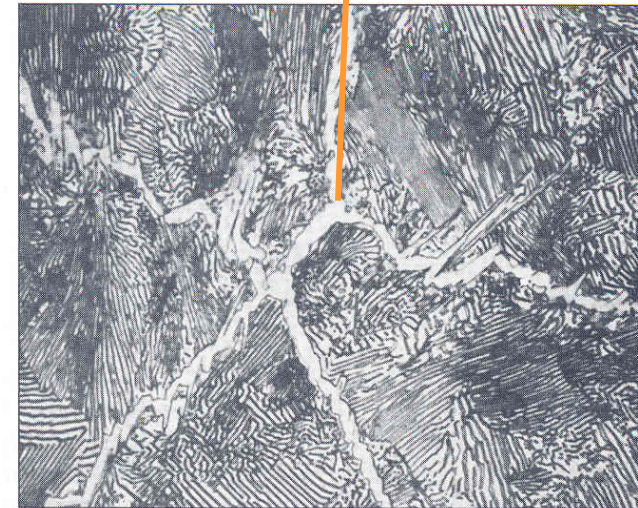


Slow Cooling of Plain Carbon Steel



cementite

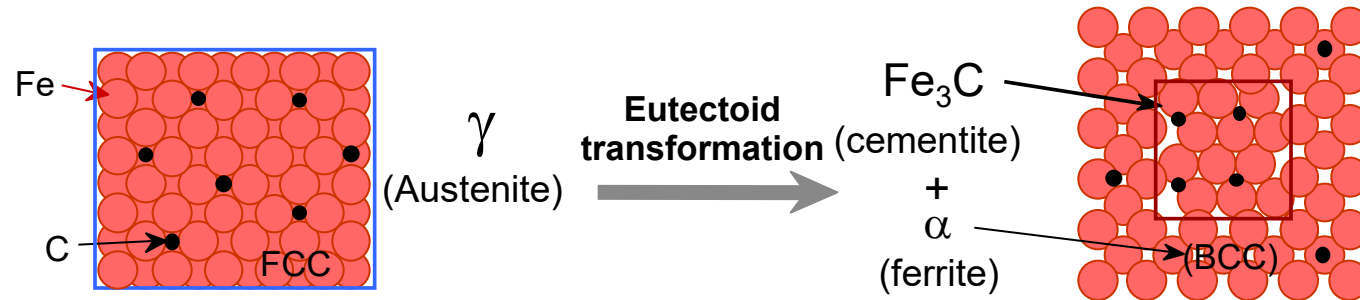
Fe_3C



Phase Transformations

Slow Cooling, why?

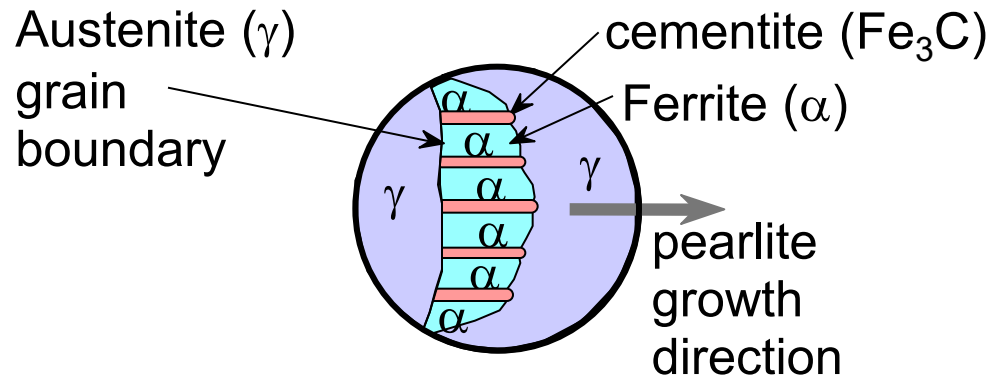
- It takes very long time for one phase to transform into another



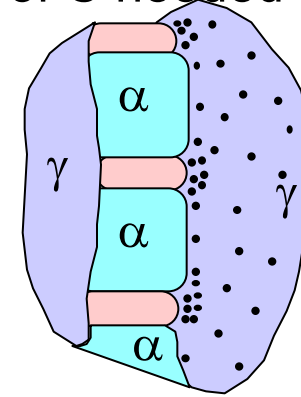
- The rate of transformation vary with time and T .
- The mechanical properties of Materials depend on non-equilibrium structures.

Eutectoid Transformation Rate

- Growth of pearlite from austenite:



Diffusive flow
of C needed

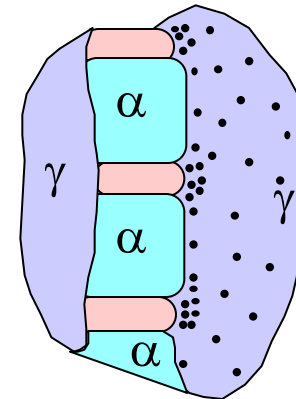
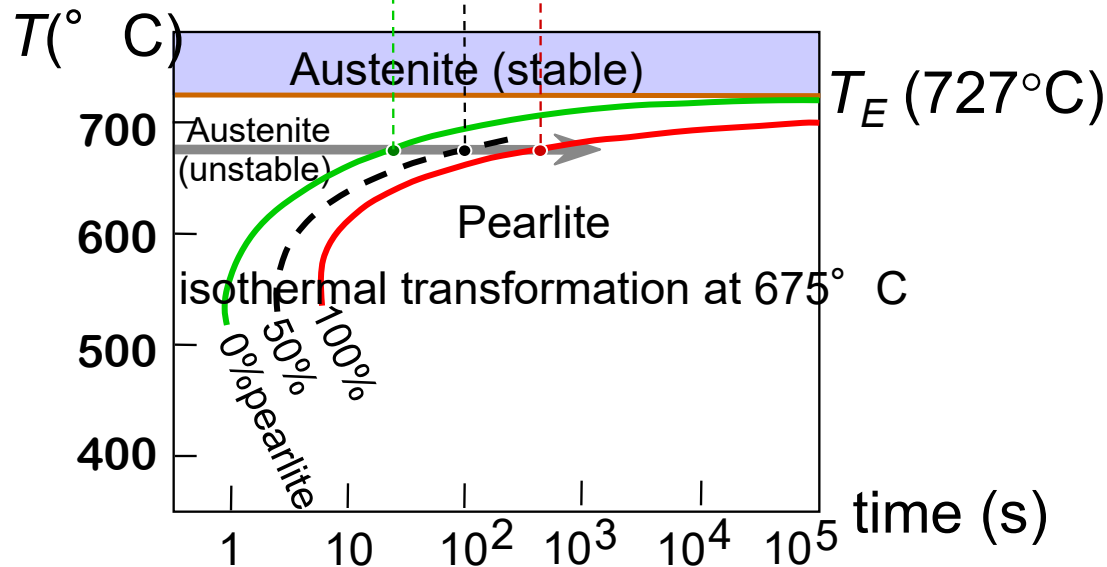
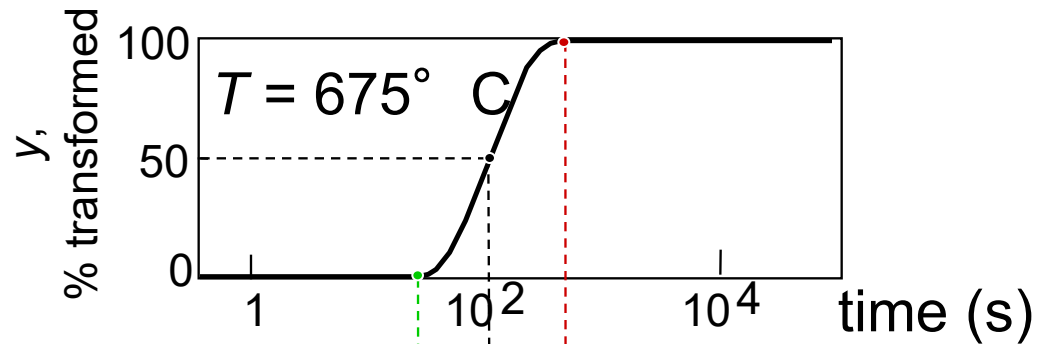


Course pearlite $\rightarrow$ formed at higher T - softer

Fine pearlite $\rightarrow$ formed at low T - harder

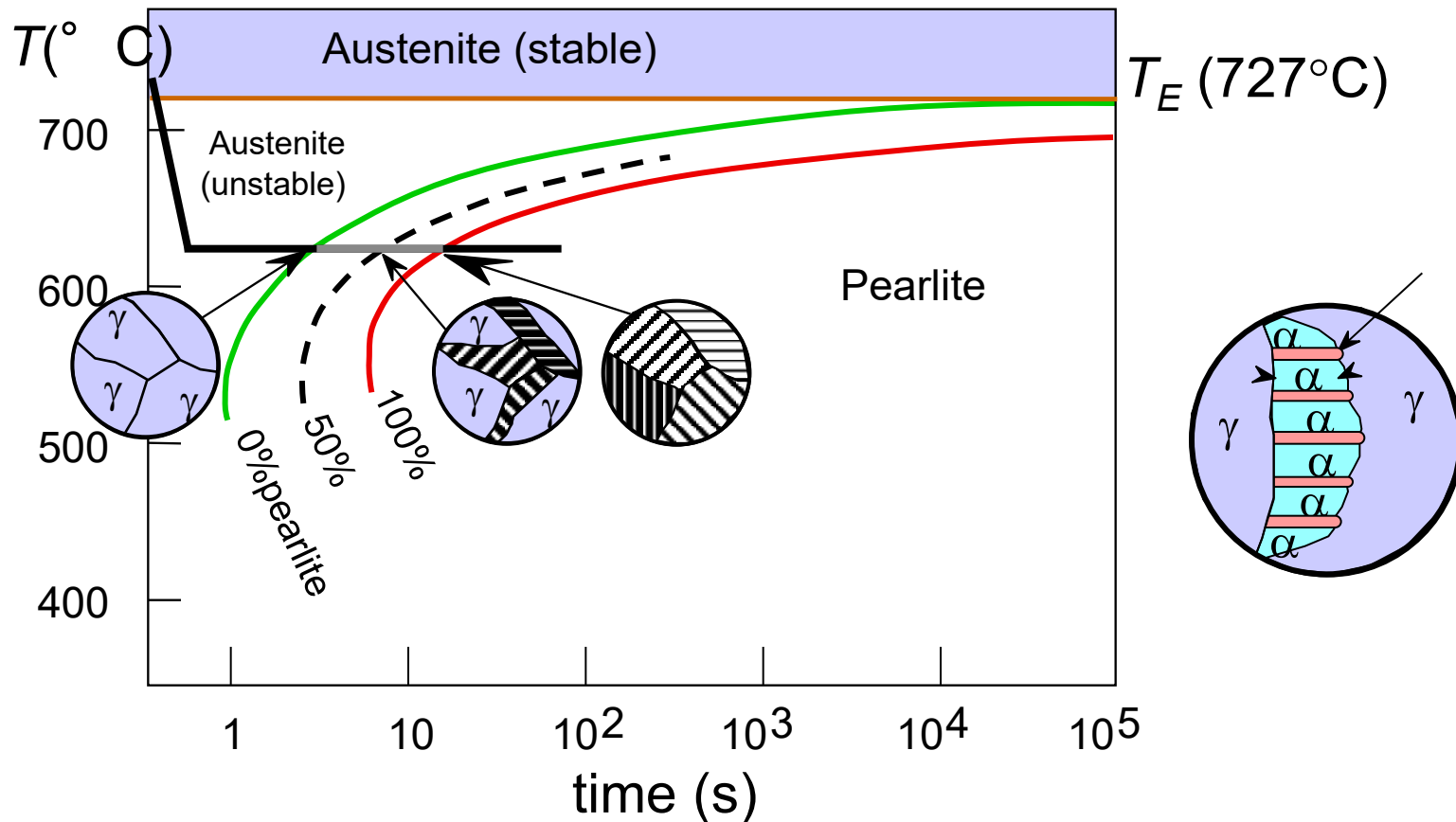
Isothermal Transformation Diagrams

- Fe-C system, $C_o = 0.76 \text{ wt\% C}$
- Transformation at $T = 675^\circ \text{ C}$.



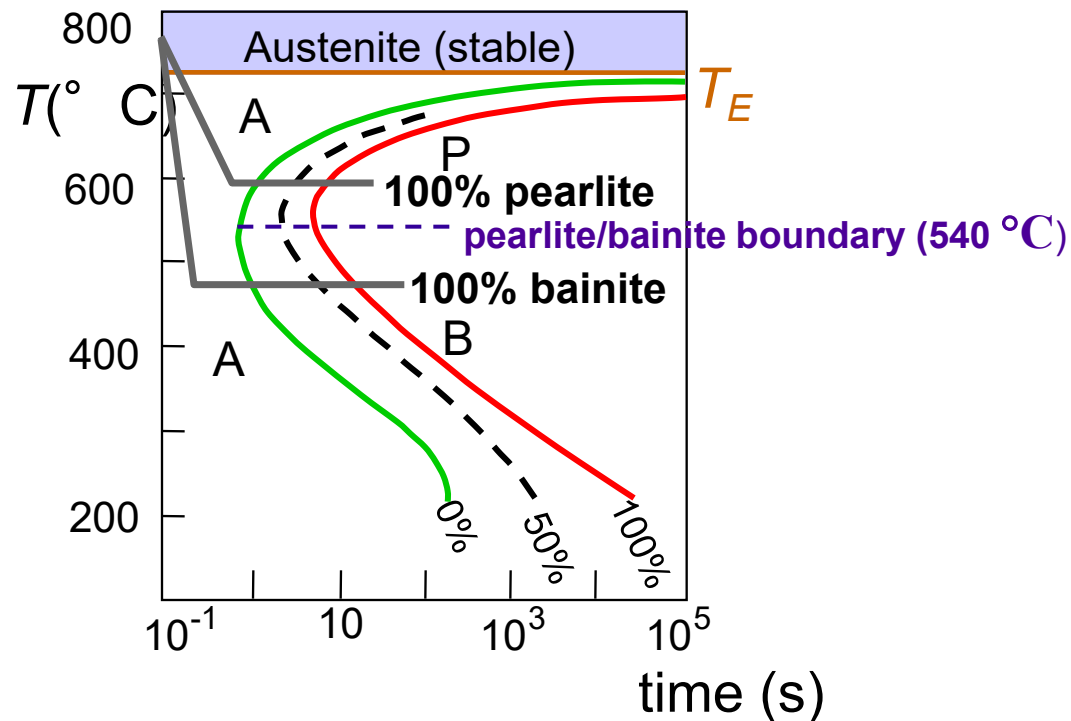
Effect of Cooling History in Fe-C System

- Eutectoid composition, $C_0 = 0.76 \text{ wt\% C}$
- Begin at $T > 727^\circ \text{ C}$
- Rapidly cool to 625° C and hold isothermally.

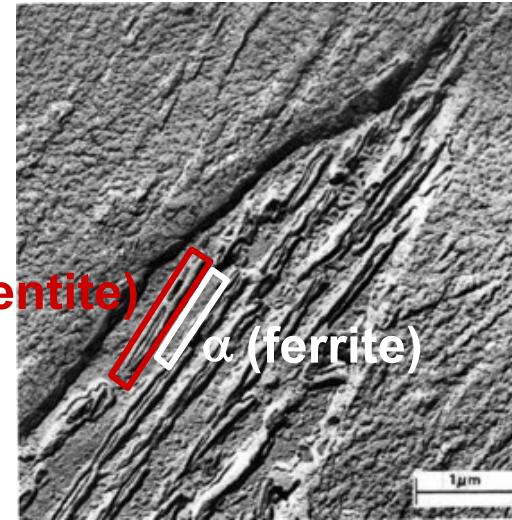


Non-Equilibrium Transformation Products: Fe-C

- Bainite (Below 540 °C):
 - α strips with long rods of Fe_3C
 - diffusion controlled.
- Isothermal Transf. Diagram

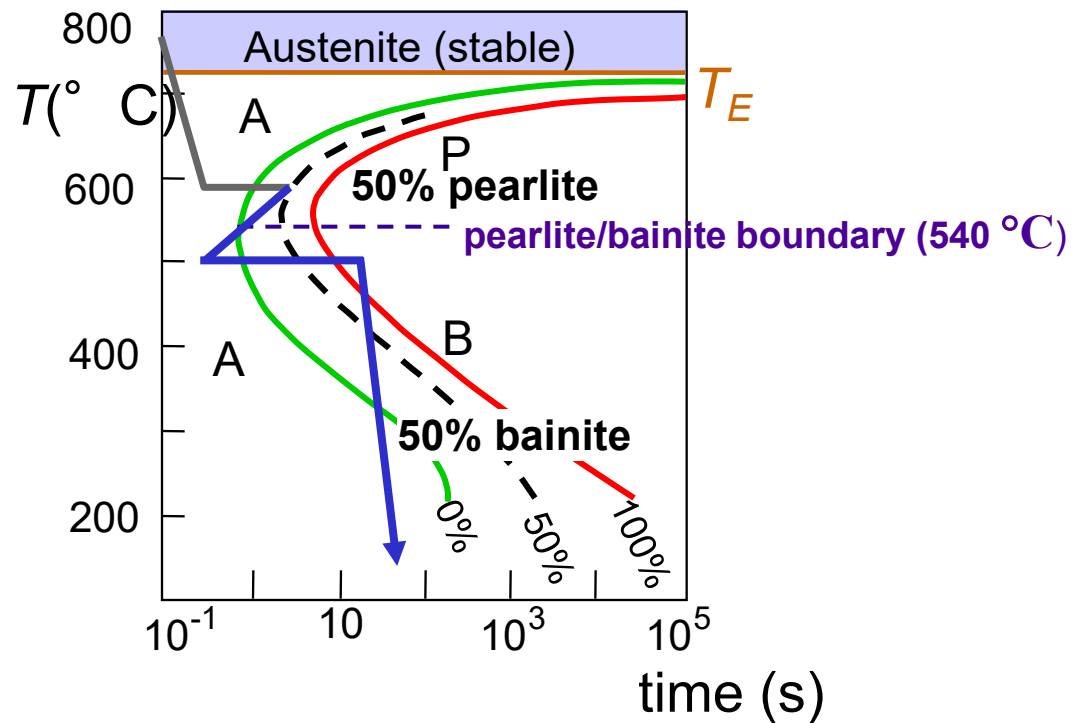


Fe_3C
(cementite)



5 μm

50% Pearlite and 50% Bainite?



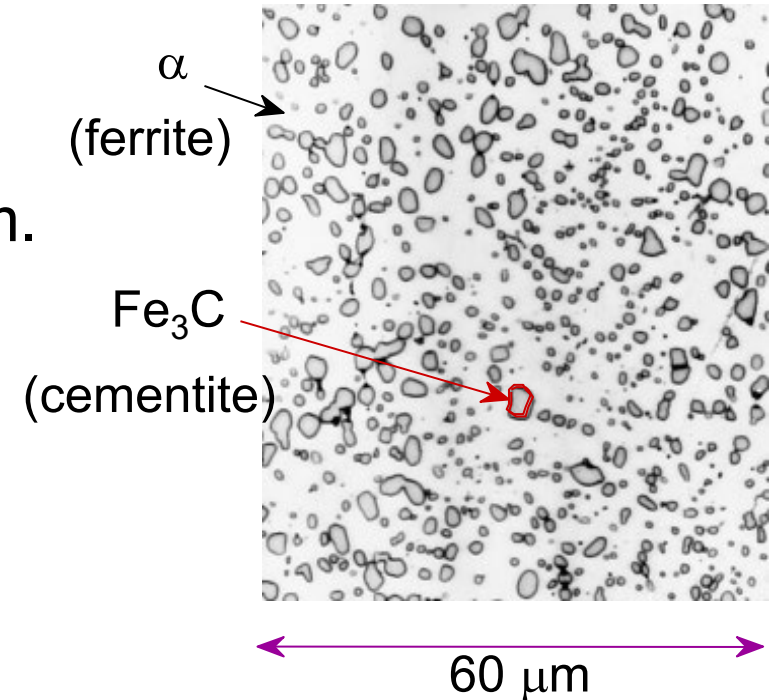
Spheroidite: Fe-C System

Spheroidite:

If pearlite or bainite microstructures are heated to a temperature below the eutectoid temperature (727 °C) and kept for a sufficient long time, spheroidite will form.

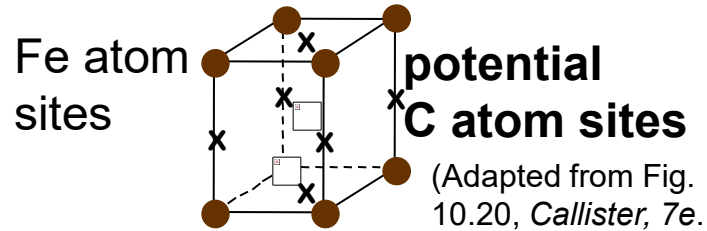
- α grains with spherical Fe_3C
- diffusion dependent.
- reduces interface

Soft and ductile

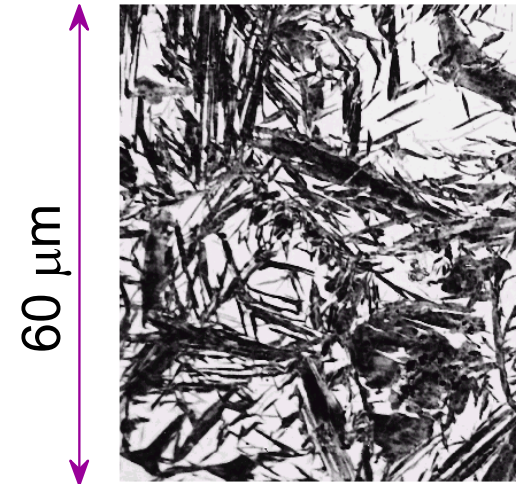
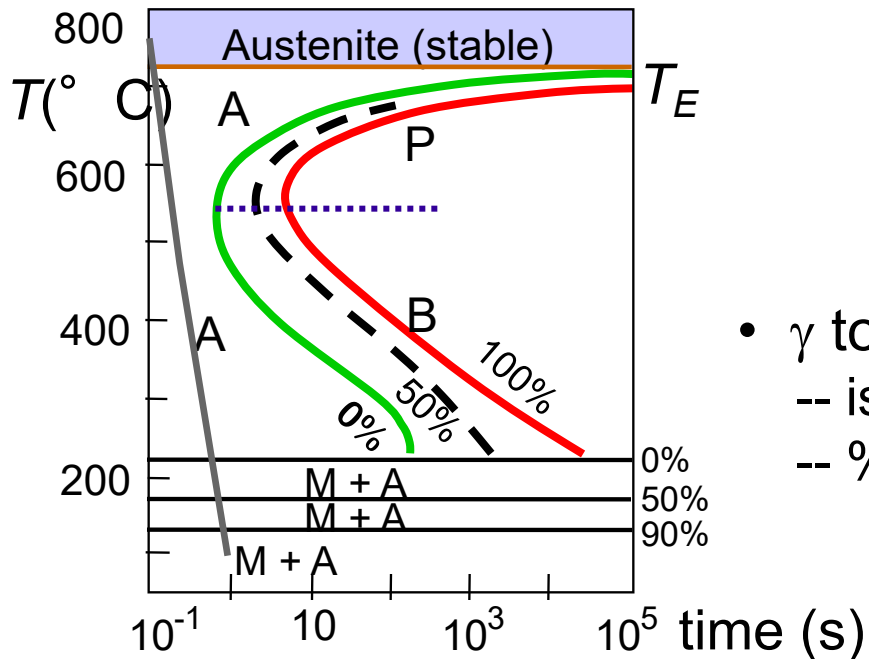


Martensite: Fe-C System

- Martensite:
-- γ (FCC) to Martensite (BCT)
(involves single atom jumps)



- Isothermal Transf. Diagram

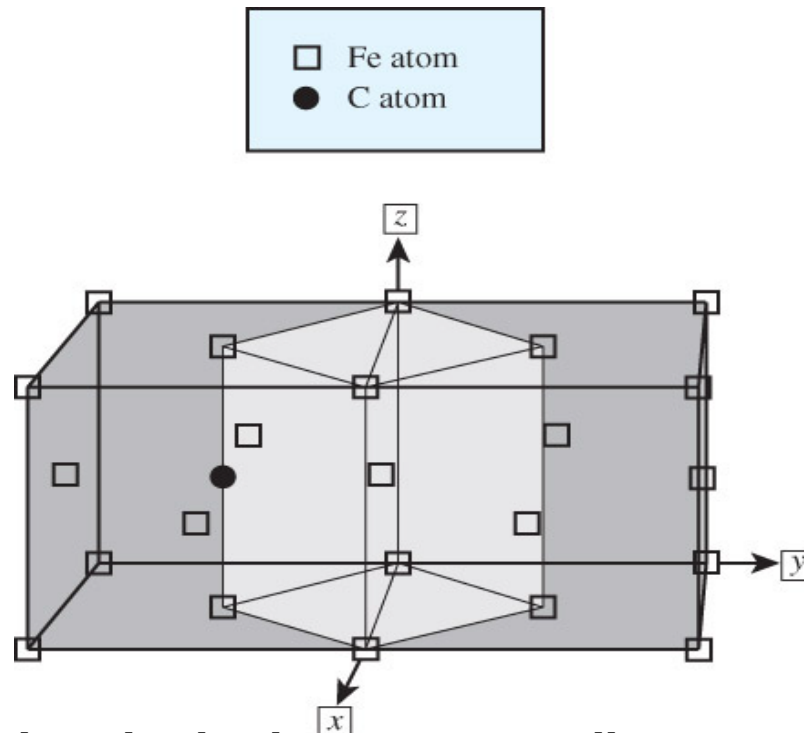


— Martensite needles
— Austenite

- γ to M transformation..
-- is rapid!
-- % transf. depends on T only.

The Martensite Transformation

- presence of interstitial carbon atoms in iron that leads to the tremendous lattice strain during the diffusionless transformation of austenite during quenching (i.e., quench hardening).



How NASA Reinvented The Wheel - Shape Memory Alloys

<https://www.youtube.com/watch?v=2lv6Vs12jLc&feature=youtu.be>

REAL
ENGINEERING

HOW NASA REINVENTED THE WHEEL

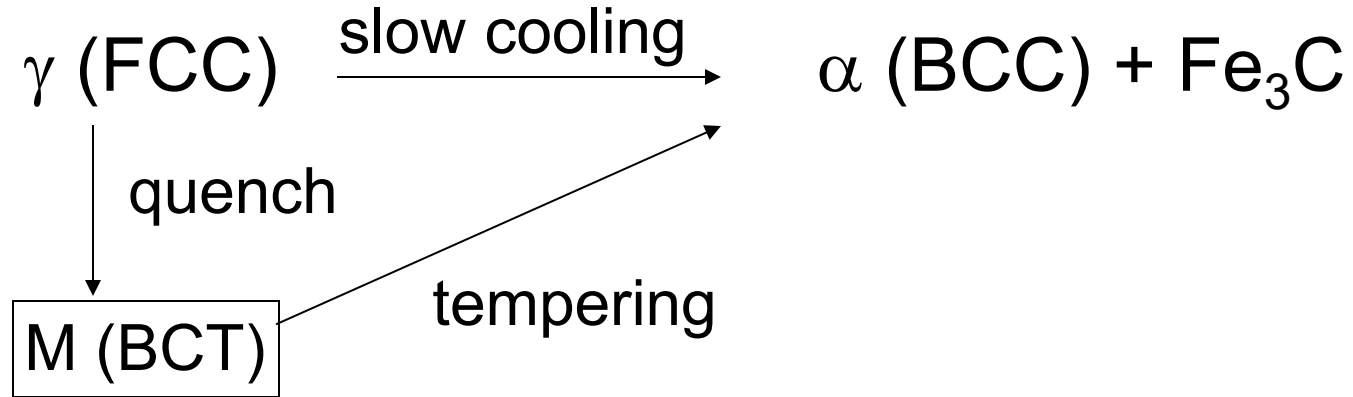


Diversity of Properties for Steels

Table 13.2 Approximate Tensile Strength and Ductility Properties for Various Phases and Microstructural Constituents Found in Heat-Treated Steels (including hot-rolled steels)

Phase or constituent	Yield strength (MPa/ksi)	Tensile strength (MPa/ksi)	Ductility (% elongation)	Hardness (BHN/HRC)
Pure iron	~100/~15	~172/~25	~60	~70/NA
Ferrite	~172/~25	~241/~35	~40	95–100/NA
Cementite	–	34T, 380C/5T, 200C	0	>750/>70
Coarse pearlite	207–310/30–45	241–448/35–65	30–20	100–125/<5
Medium pearlite	310–414/45–60	448–621/65–90	20–12	125–160/5–20
Fine pearlite	414–621/60–90	552–793/80–115	12–5	160–250/20–35
Upper bainite	621–828/90–120	828–1103/120–160	12–7	300–420/35–45
Lower bainite	828–1034/120–150	1034–1517 150–220	7–3	420–550/45–55
As-quenched martensite	1034–1586/150–230+	1172–1724/170–250+	0	320–740/35–65+
Tempered martensite	862–1379/125–200	965–1655/140–240	5–1	285–580/30–55
Low-carbon steel (<0.25%)	172–310/25–45	207–483/30–70	40–27	95–140/<10
Medium-carbon steel (0.25–0.45%)	310–414/45–60	483–655/70–95	27–15	140–190/10–25
High-carbon steel (0.45–1.2%)	414–551/60–80	655–966/95–140	15–2	285–580/30–55

Martensite Formation and Tempering

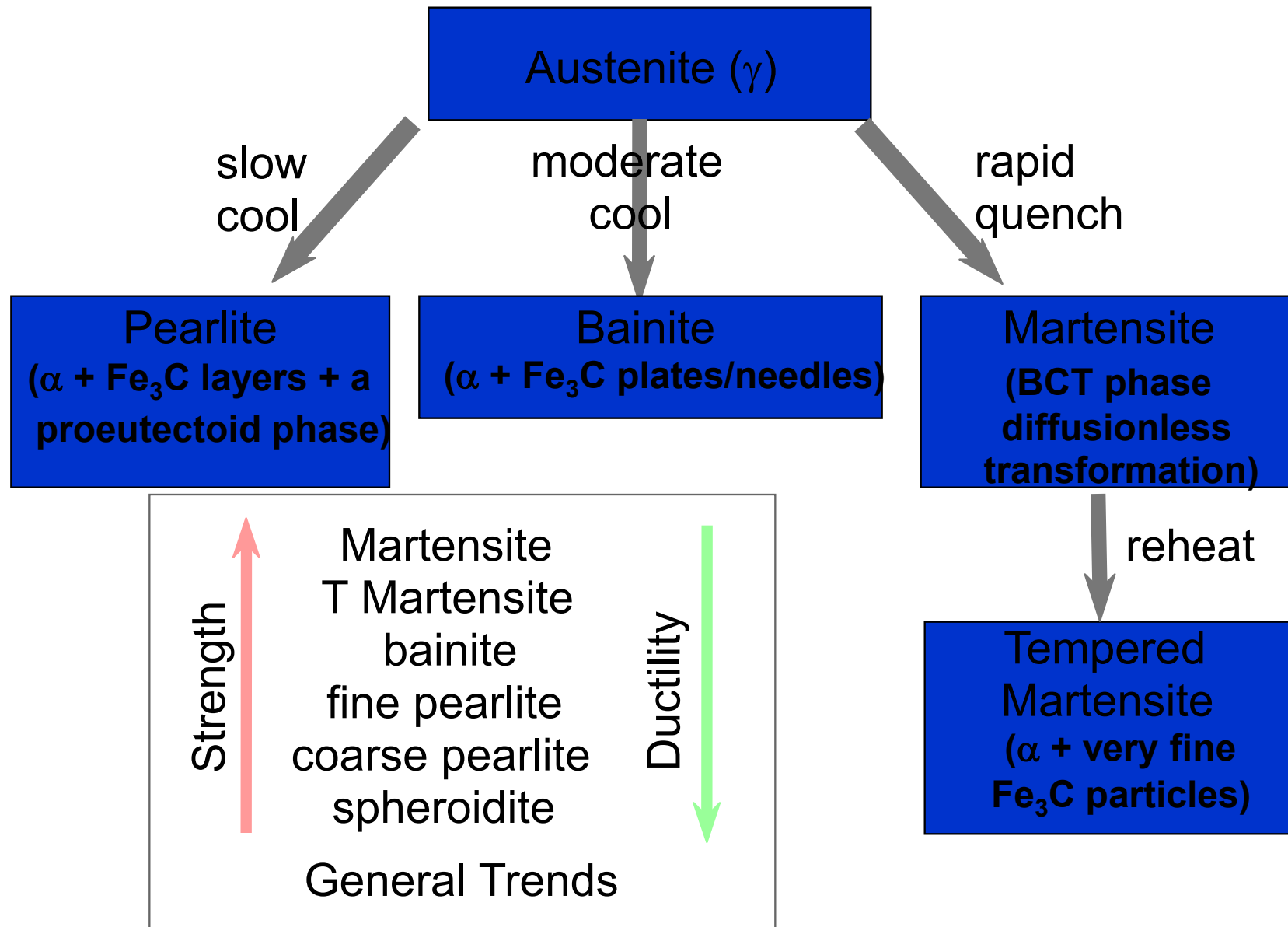


M = martensite is body centered tetragonal (BCT)

Tempering: Reheat martensite to 250 - 650°C

Excess carbon precipitates out to form fine particles of Fe_3C

Processing Options



Continuous Cooling Transformation of Steels

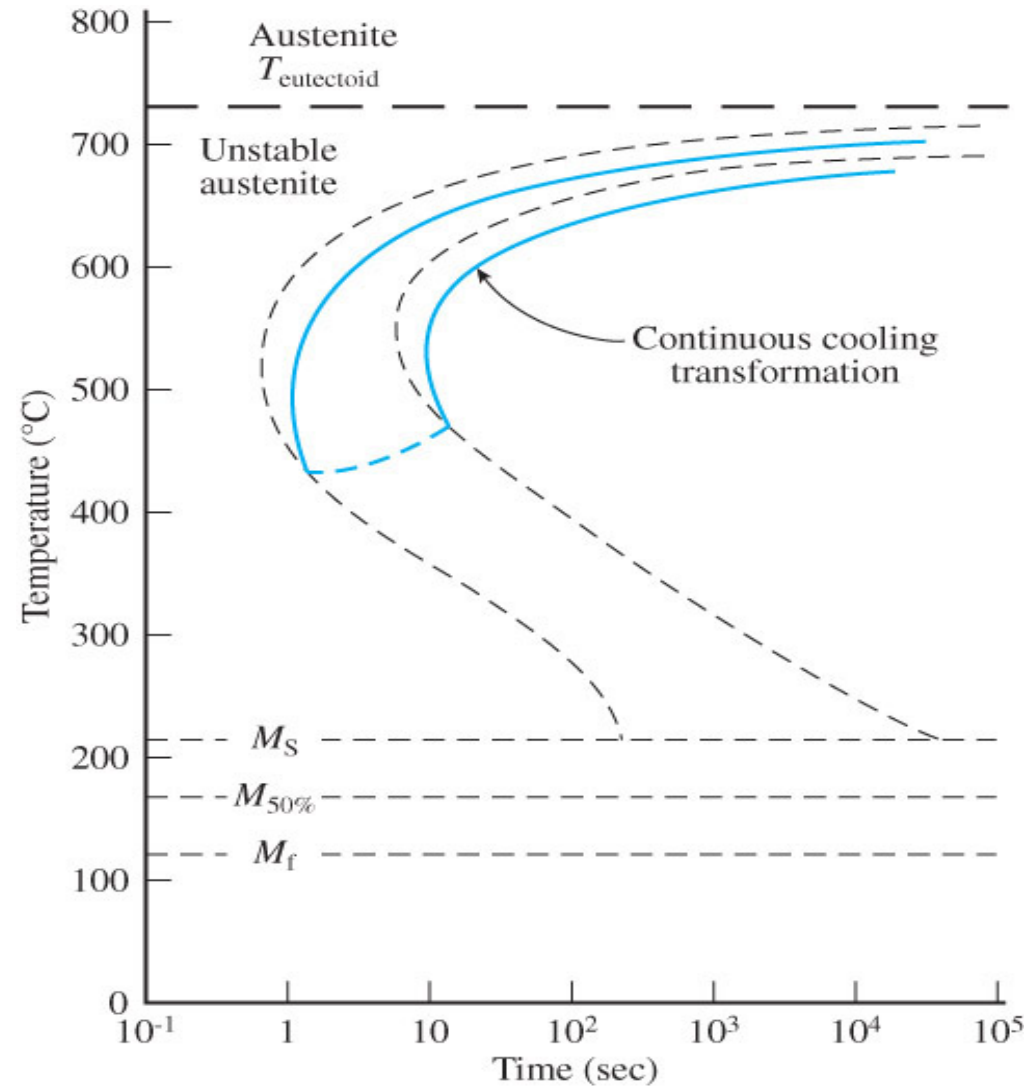
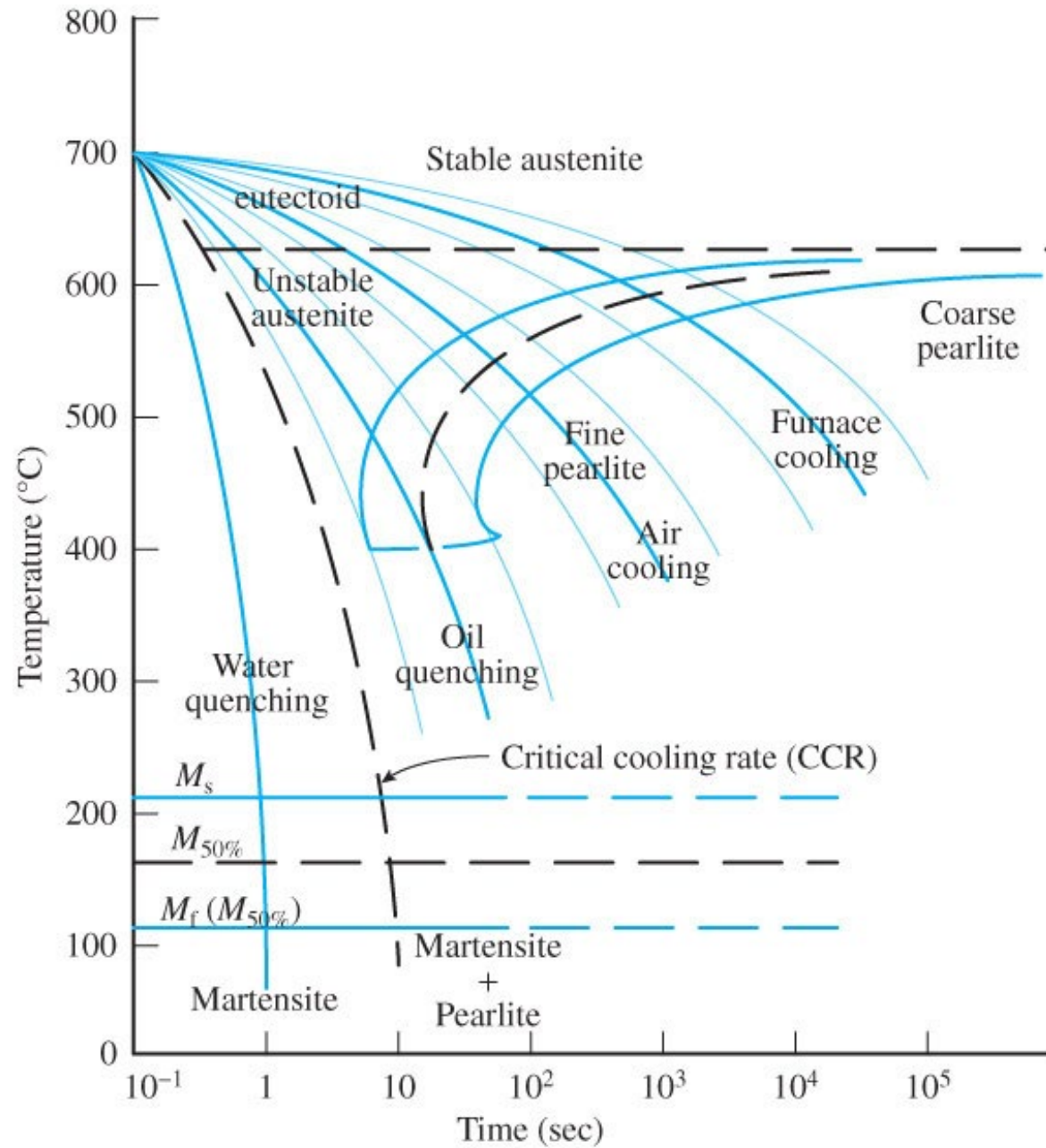


Figure 13.30



**The Complete
CCT Diagram for
a Eutectoid Steel**



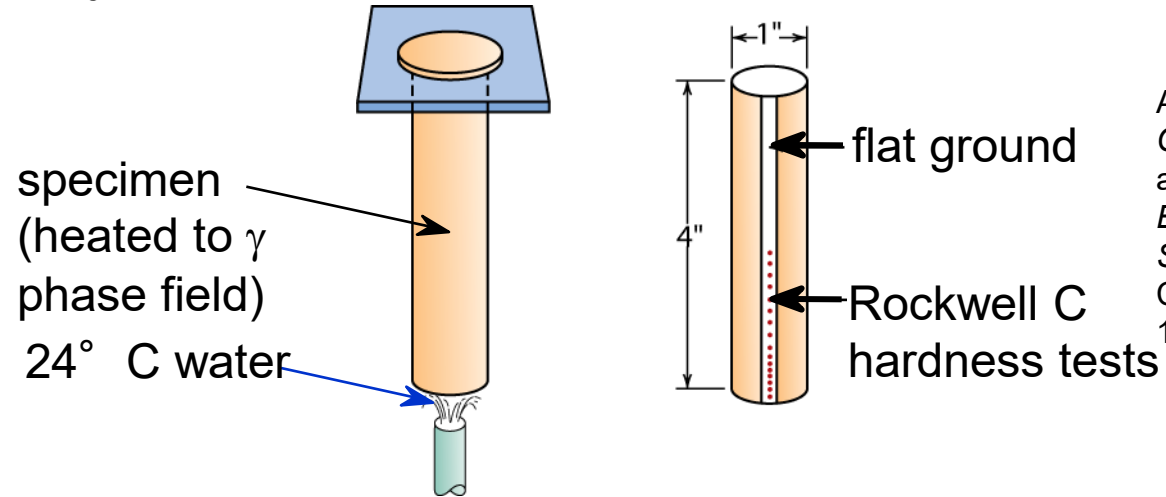
<https://www.youtube.com/watch?v=mYsFUFgOEmM>

"Born of Fire", also known as "Imported from Detroit", is the television commercial that introduced the Chrysler 200 to the public. The commercial ran for the first time during Super Bowl XLV, and, at two minutes long, it is one of the longest commercials ever shown on such event.

- Release date(s): [February 6, 2011](#)
- Running time: 2:02
- CBS is selling 30-second Super Bowl 2019 commercial spots to the tune of \$5.1 to \$5.3 million, as Bloomberg first reported.

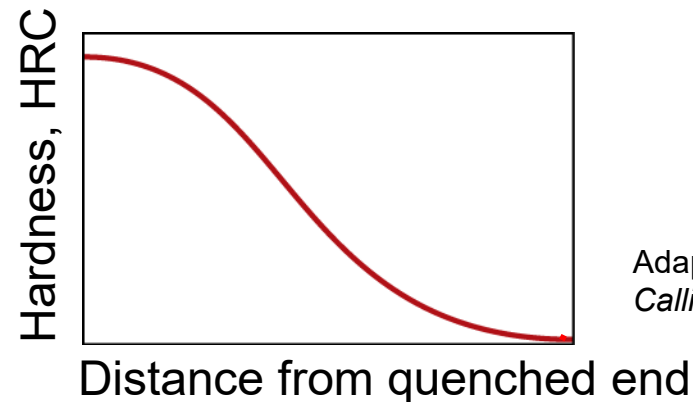
Hardenability--Steels

- Ability to form martensite
- Jominy end quench test to measure hardenability.



Adapted from Fig. 11.11, *Callister 7e*. (Fig. 11.11 adapted from A.G. Guy, *Essentials of Materials Science*, McGraw-Hill Book Company, New York, 1978.)

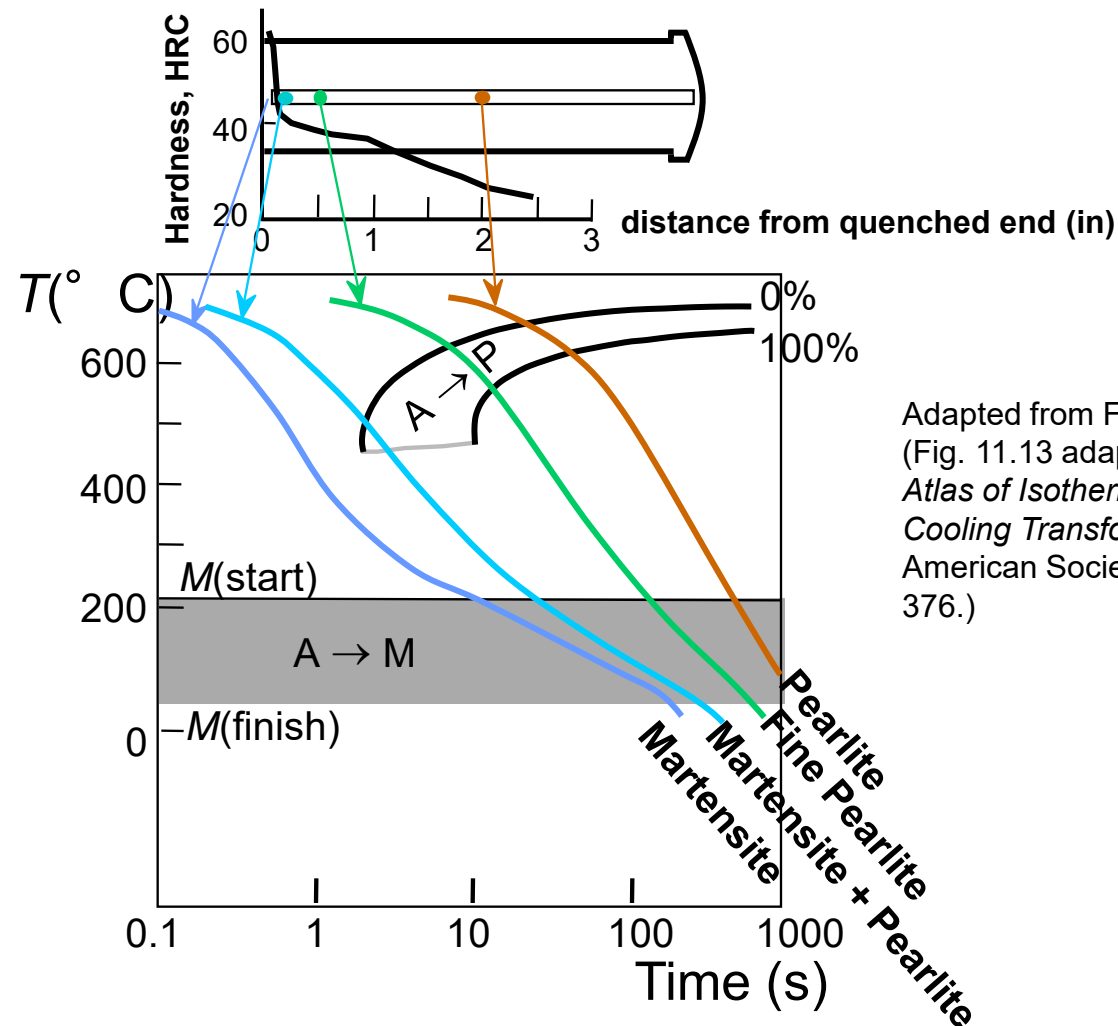
- Hardness versus distance from the quenched end.



Adapted from Fig. 11.12, *Callister 7e*.

Hardness Changes W/Position

- The cooling rate varies with position.



Adapted from Fig. 11.13, *Callister 7e*.
(Fig. 11.13 adapted from H. Boyer (Ed.)
*Atlas of Isothermal Transformation and
Cooling Transformation Diagrams*,
American Society for Metals, 1977, p.
376.)

Quenching Medium & Geometry

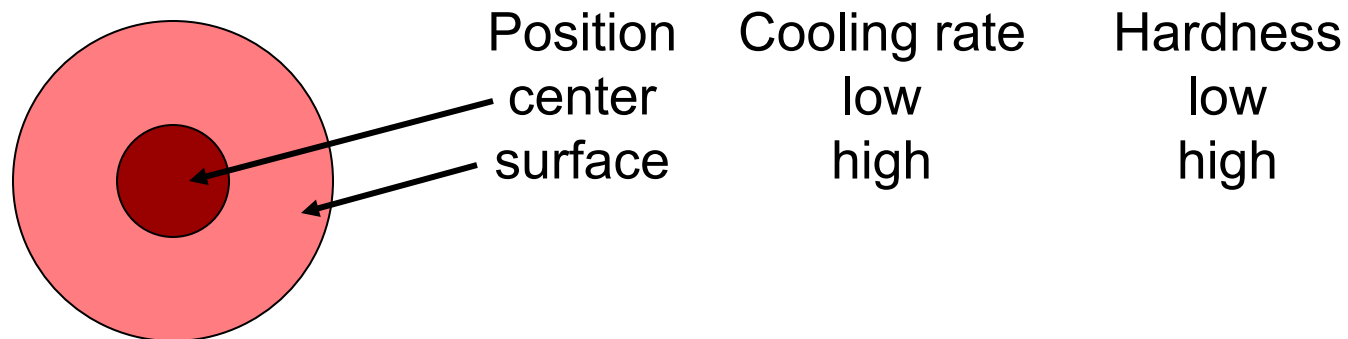
- Effect of quenching medium:

Medium	Severity of Quench	Hardness
air	low	low
oil	moderate	moderate
water	high	high

- Effect of geometry:

When surface-to-volume ratio increases:

- cooling rate increases
- hardness increases



Hard Turning - Background

- Cutting hard materials is difficult
- Grinding, the most popular method, is expensive, environmentally unfriendly, and inflexible
- Diamond and CBN (Cubic boron nitride) tools appear
- Turning hard material is possible. Hard turning is a technology that may replace grinding
- Lathe, no cutting fluid, and flexible

https://www.youtube.com/watch?v=tK-gD_DPWzE

Miyano GN-3200 Hard Turning Lathe



White Layer Formation, Why?

- White layer is easy to peel off and is harmful to product life

